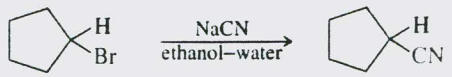
Organic Cume November 11, 2010

The following questions were taken from the undergraduate textbook *Organic Chemistry* by Carey and Giuliano, 8th Ed.

Provide the mechanism, sequence, explanation or product that best answers the question.

8.47 The reaction of cyclopentyl bromide with sodium cyanide to give cyclopentyl cyanide



Cyclopentyl bromide Cyclopentyl cyanide

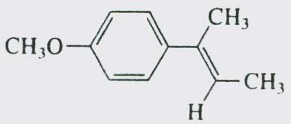
proceeds faster if a small amount of sodium iodide is added to the reaction mixture. Can you suggest a reasonable mechanism to explain the catalytic function of sodium iodide?

10.45 Alkenes slowly undergo a reaction in air called *autoxidation* in which allylic hydroperoxides are formed.

Cyclohexene Oxygen 3-Hydroperoxycyclohexene

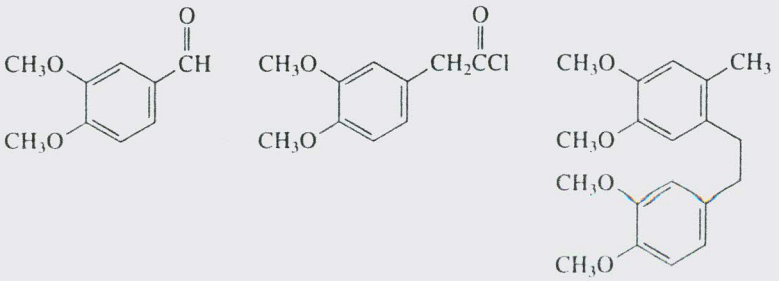
Keeping in mind that oxygen has two unpaired electrons (·Ö:Ö·), suggest a reasonable mechanism for this reaction.

11.42 Hydroboration–oxidation of (*E*)-2-(*p*-anisyl)-2-butene yielded an alcohol A, mp 60°C, in 72% yield. When the same reaction was performed on the *Z* alkene, an isomeric liquid alcohol B was obtained in 77% yield. Suggest reasonable structures for A and B, and describe the relationship between them.



(*E*)-2-(*p*-Anisyl)-2-butene

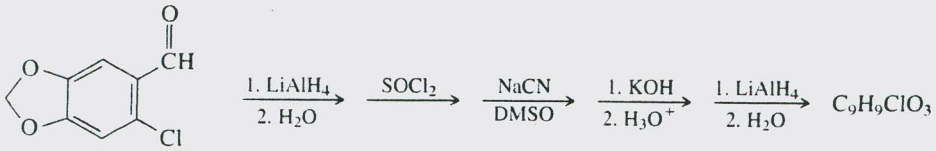
12.56 The synthesis of compound C was achieved by using compounds A and B as the source of all carbon atoms. Suggest a synthetic sequence involving no more than three steps by which A and B may be converted to C.



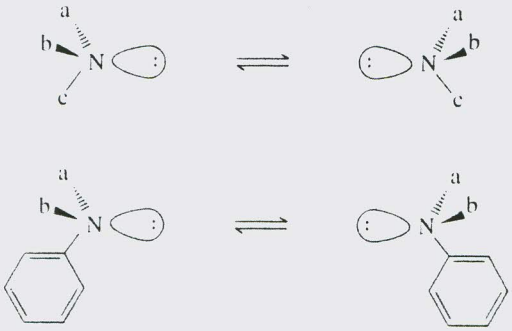
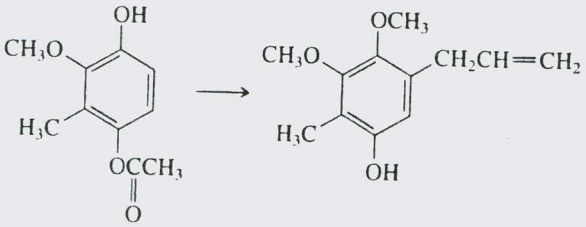
Compound A Compound B Compound C

16.36 The *p*-toluenesulfonate shown undergoes an intramolecular Williamson reaction on treatment with base to give a spirocyclic ether. Demonstrate your understanding of the terminology used in the preceding sentence by writing the structure, including stereochemistry, of the product.

18.22 The compound shown was subjected to the following series of reactions to give a product having the molecular formula $C_9H_9ClO_3$. What is this product?



21.32 Both alkyl- and arylamines have a low barrier for inversion at nitrogen, which prevents the separation of chiral amines into their enantiomers. The barrier for inversion at nitrogen in alkylamines is approximately 25 kJ/mol (6 kcal/mol), whereas for arylamines it is much lower, on the order of 6.3 kJ/mol (1.5 kcal/mol). Can you suggest a reason for the difference?



22.26 Describe a scheme for carrying out the following synthesis. (In the synthesis reported in the literature, four separate operations were required.)